AIDCF

AI Data Centre Facilities Assurance Framework

ASSURANCE PRINCIPLES STANDARD

Version 1.0

The constitutional rules governing claims, evidence, uncertainty, critical gates, decisions, independence and responsible use

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Assurance is bounded confidence supported by evidence, not confidence created by presentation.

Document control

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Executive thesis

AIDCF assurance is a bounded, evidence-supported and accountable conclusion about whether the facilities and infrastructure supporting a defined AI Capability Instance can be reasonably relied upon to achieve required Service Outcomes within declared boundaries, states, operating conditions and time. It is not a guarantee, a generic quality label, a score, a maturity level or an inherited certificate.

Constitutional reasoning chain

Decision Question → Capability Instance → Service Outcomes → Boundary and Operating Envelope → Assets and Dependencies → Adverse Conditions and Failure Paths → Safeguards and Recovery → Evidence and UNKNOWN → Critical Gates → Independent Review → Bounded Assurance Decision

Contents

1. Authority, scope and constitutional status
2. Assurance architecture and terminology
3. Capability and outcome principles
4. Scope, boundary and state principles
5. Dependency and concentration principles
6. Failure, resilience and recovery principles
7. Safeguard and validation principles
8. Evidence and confidence principles
9. UNKNOWN and limitation principles
10. Critical-gate principles
11. Cross-domain and graph-native principles
12. Decision, scoring, maturity and reporting principles
13. Independence, competence and ethics
14. Lifecycle and continuing assurance
15. Claim language and decision model
16. Conformance and enforcement
17. Machine-readable profile
18. Worked applications and anti-patterns
19. Appendices

Part I - Authority, scope and constitutional status

1. Purpose

This standard establishes the non-negotiable principles that constrain how AIDCF requirements are designed, assessments are performed, evidence is interpreted and conclusions are issued. Downstream artifacts may operationalise these principles but shall not weaken, bypass or silently reinterpret them.

2. Artifact precedence

Layer	Authority
Manifesto	Purpose, identity and public commitments.
Core Conceptual Model	Canonical meanings and reasoning.
Artifacts 003-005	Ontology, failure semantics and graph integration.
Artifact 006	Constitutional assurance rules.
Downstream standards	Operational controls, methods, scores, reports and schemes.
Tooling	Declared implementation only, never hidden authority.

3. Constitutional invariants

ID	Invariant
CI-01	Capability and Service Outcome are the primary assurance object.
CI-02	Assertions do not become facts without evidence and review.
CI-03	UNKNOWN remains visible and non-favourable.
CI-04	Critical gates cannot be averaged away.
CI-05	Cross-boundary dependencies remain visible.
CI-06	Designed, installed, commissioned, observed and tested states remain distinct.
CI-07	Conclusions stay bounded by scope, state, time and evidence.
CI-08	Material change can invalidate a prior conclusion.
CI-09	Automation proposes; accountable authority approves.
CI-10	AIDCF and AI Trust Graph retain separate native conclusions.

4. Non-substitution

- Does not replace licensed engineering or statutory approval.
- Does not determine legal compliance unless separately authorised.
- Does not authorise intrusive or hazardous testing.
- Does not transfer operational accountability from owners to assessors.

Part II - Assurance architecture and terminology

5. Definition of assurance

Assurance is a reasoned and reviewable level of confidence, for a defined decision, that specified claims about a Capability Instance and supporting infrastructure are supported by sufficient appropriate evidence within a declared scope, state, period and set of limitations.

6. Assurance levels

Level	Permitted conclusion	Minimum basis
A0 No assurance	No positive conclusion.	No/insufficient evidence or critical uncertainty.
A1 Descriptive	Known state described.	Traceable inventory and observations.
A2 Design	Requirements/design supported.	Approved design evidence and competent review.
A3 Implementation	Measures implemented in observed scope.	Corroborated technical/as-built evidence.
A4 Operating	Measures effective for tested conditions/period.	Representative operating/test evidence.
A5 Resilience	Capability sustained or recovered for defined scenarios.	Cross-domain scenario, gate and recovery evidence.

7. Mandatory distinctions

Do not collapse	Reason
Assurance / guarantee	Assurance is bounded confidence, not certainty.
Evidence grade / confidence	Support strength differs from reviewer confidence.
Control / safeguard	Requirement differs from real-world mechanism.
Maturity / effectiveness	Institutionalisation differs from performance.
Score / decision	Number cannot replace reasoning and gates.
UNKNOWN / failure	Missing knowledge differs from demonstrated adverse state.
Not Applicable / not tested	Irrelevance differs from missing activity.
Failover / recovery	Transfer differs from restoration.
Redundancy / resilience	Duplication does not prove outcome continuity.
Connection / authority	Reachability does not prove permission or consequential action.

Part III - Capability and outcome principles

APS-001 Capability-first assurance

The primary object of AIDCF assurance shall be a defined Capability Instance and required Service Outcomes.

Assets and controls matter because of the functions and outcomes they support.

Dimension	Requirement
Application	Define capability, owner, architecture baseline, lifecycle stage, outcomes and criticality.
Prohibited interpretation	Generic facility assurance without a Capability Instance.
Minimum evidence	Capability record, outcome requirements and architecture.
Decision effect	Missing capability or outcome blocks positive capability assurance.

APS-002 Outcome alignment

Every material test, control, finding and recommendation shall identify the Service Outcome or decision it protects.

Outcome alignment prevents checklist completion from becoming the objective.

Dimension	Requirement
Application	Record affected outcome, tolerance, time sensitivity and consequence.
Prohibited interpretation	Control activity justified only because it is customary.
Minimum evidence	Service requirements and impact analysis.
Decision effect	Unlinked activity cannot determine capability assurance.

APS-003 Function before product

Required functions shall be defined independently from products implementing them.

This preserves vendor neutrality and enables substitution and failure analysis.

Dimension	Requirement
Application	Define function, demand, duration, state and implementing assets.
Prohibited interpretation	Using a vendor product name as the control objective.
Minimum evidence	Functional requirements and observed implementation.
Decision effect	Unclear function limits design and effectiveness conclusions.

APS-004 Outcome-centric materiality

Materiality shall be based on plausible effect on outcomes, safety, stakeholders and obligations.

Issue count, novelty and cost alone do not establish materiality.

Dimension	Requirement
Application	Assess magnitude, duration, concentration, recoverability and stakeholder effect.
Prohibited interpretation	Averaging catastrophic and trivial effects.
Minimum evidence	Capability impact analysis and accountable criteria.
Decision effect	Materiality determines depth, gates and escalation.

Part IV - Scope, boundary and state principles

APS-005 Explicit boundary

Every conclusion shall declare included systems, locations, organisations, states and time.

A boundary makes claims interpretable.

Dimension	Requirement
Application	Record inclusions, exclusions and boundary-crossing dependencies.
Prohibited interpretation	Treating out-of-scope as non-existent.
Minimum evidence	Approved scope, architecture and dependency record.
Decision effect	Unclear boundary blocks final assurance wording.

APS-006 Operating-envelope discipline

Assurance applies only within defined load, environmental, capacity, duration and state conditions.

Evidence in one condition cannot automatically support another.

Dimension	Requirement
Application	State demand, headroom, limits, duration and applicable states.
Prohibited interpretation	Generalising low-load tests to peak AI load.
Minimum evidence	Operating envelope, telemetry and test conditions.
Decision effect	Claims outside the envelope remain unassured.

APS-007 State-specific assurance

Normal, degraded, maintenance, emergency, recovery and transition states shall be considered separately where material.

Normal operation can hide maintenance or recovery fragility.

Dimension	Requirement
Application	Identify state coverage for each claim and safeguard.
Prohibited interpretation	Assuming normal-state evidence proves all states.
Minimum evidence	State model, work controls, incidents and tests.
Decision effect	Uncovered critical states cap resilience assurance.

APS-008 Reality over intent

Observed and tested reality shall take precedence over intended design where they conflict.

Intent cannot prove current implementation.

Dimension	Requirement
Application	Separate intended, designed, installed, commissioned, observed and tested state.

Dimension	Requirement
Prohibited interpretation	Silently reconciling records to match design.
Minimum evidence	Current configuration, observation and test evidence.
Decision effect	Conflict is disclosed and evaluated.

APS-009 Proportionate depth

Assessment depth shall match consequence, complexity, change, uncertainty and decision need.

Proportionality prevents both superficiality and unsafe overtesting.

Dimension	Requirement
Application	Document depth, sampling and escalation rationale.
Prohibited interpretation	One fixed checklist depth for all capabilities.
Minimum evidence	Scoping, criticality and test authorisation.
Decision effect	Insufficient depth limits the conclusion.

Part V - Dependency and concentration principles

APS-010 Dependency visibility

Material dependencies shall be explicit, typed, directional and state-aware.

Hidden dependencies undermine redundancy and recovery claims.

Dimension	Requirement
Application	Map upstream, downstream, shared, external, human, control and recovery dependencies.
Prohibited interpretation	Omitting outsourced dependencies.
Minimum evidence	Dependency graph, architecture, contracts and observed paths.
Decision effect	Critical hidden dependency may trigger a gate.

APS-011 External dependency continuity

External providers remain part of capability reasoning even when outside direct assessment.

Exclusion does not remove consequence.

Dimension	Requirement
Application	Identify provider, service, capacity, evidence access, failure effect and escalation.
Prohibited interpretation	Assuming an SLA proves continuity.
Minimum evidence	Contracts, provider evidence and scenario analysis.
Decision effect	Weakly evidenced critical external dependency caps assurance.

APS-012 Shared-dependency scrutiny

Nominal alternatives shall be examined for common causes, routes, controllers, identities, teams and providers.

Duplication does not establish independence.

Dimension	Requirement
Application	Map shared fault domains and common-mode events.
Prohibited interpretation	Calling dual paths resilient without shared-dependency analysis.
Minimum evidence	Topology, configuration and test evidence.
Decision effect	Unresolved common mode prevents independence claims.

APS-013 Concentration awareness

Capability, capacity, authority, supplier and knowledge concentration shall be explicit.

Concentration magnifies consequence.

Dimension	Requirement
Application	Identify aggregation points, blast radius and recovery options.

Dimension	Requirement
Prohibited interpretation	Dismissing concentration using component reliability alone.
Minimum evidence	Capacity, topology, role and provider data.
Decision effect	Critical concentration increases scenario priority.

APS-014 Dependency ownership

Every material dependency shall have an owner or explicit UNKNOWN owner.

Responsibility is required for evidence, treatment and recovery.

Dimension	Requirement
Application	Assign owner, operator, provider, maintainer and escalation.
Prohibited interpretation	Treating contract assignment as proof of control.
Minimum evidence	Responsibility model and operating procedure.
Decision effect	Unowned critical dependency is a responsibility gap.

Part VI - Failure, resilience and recovery principles

APS-015 Failure-chain analysis

Material failures shall be analysed from initiating condition through mode, effect, propagation, consequence, safeguard and recovery.

Stopping at component failure misses capability impact.

Dimension	Requirement
Application	Apply Artifact 004 distinctions and dependency paths.
Prohibited interpretation	Collapsing cause, mode, effect and consequence.
Minimum evidence	Scenario records and failure analysis.
Decision effect	Incomplete chain limits resilience conclusions.

APS-016 Cross-domain propagation

Failure analysis shall cross domain boundaries wherever dependencies transmit effect.

Resilience emerges from interactions.

Dimension	Requirement
Application	Trace power, cooling, compute, storage, network and operations effects.
Prohibited interpretation	Siloed domain assessment as end-to-end assurance.
Minimum evidence	Cross-domain graph, telemetry and tests.
Decision effect	Unexamined material path remains UNKNOWN.

APS-017 Compound conditions

Credible combinations of failure, maintenance, demand, threat and recovery shall be considered.

Systemic failures often require multiple conditions.

Dimension	Requirement
Application	Analyse latent, sequential, compound and common-cause scenarios.
Prohibited interpretation	Dismissing combinations solely on individual rarity.
Minimum evidence	Scenario basis, incident data and stress analysis.
Decision effect	Catastrophic credible scenario may become a gate.

APS-018 Resilience lifecycle

Resilience shall cover anticipate, prevent, detect, decide, respond, contain, withstand, degrade, recover and adapt as applicable.

Prevention alone is not resilience.

Dimension	Requirement
Application	Map safeguards and gaps across the lifecycle.

Dimension	Requirement
Prohibited interpretation	Equating monitoring or redundancy with resilience.
Minimum evidence	Safeguard model, exercises and recovery records.
Decision effect	Missing essential lifecycle capability limits resilience assurance.

APS-019 Recovery evidence

Recovery claims shall identify prerequisites, sequence, authority, resources and validation.

A plan does not prove recoverability.

Dimension	Requirement
Application	Validate critical recovery steps and dependencies safely.
Prohibited interpretation	Assuming failover equals recovery.
Minimum evidence	Recovery tests, incidents and baselines.
Decision effect	Untested critical recovery remains design intent.

APS-020 Safe validation

Validation shall be authorised, competent, proportionate and non-destructive by default.

Assurance does not justify new unmanaged risk.

Dimension	Requirement
Application	Use least intrusive sufficient method, rules of engagement and abort criteria.
Prohibited interpretation	Hazardous live testing without explicit authority.
Minimum evidence	Authorisation, risk assessment, test plan and results.
Decision effect	Unsafe evidence is not accepted and is escalated.

Part VII - Safeguard and validation principles

APS-021 Safeguard reality

A control requirement and real-world safeguard shall remain distinct.

A policy cannot substitute for implemented protection.

Dimension	Requirement
Application	Identify mechanism, owner, scope, states and dependencies.
Prohibited interpretation	Counting a control statement as an operating safeguard.
Minimum evidence	Configuration and observation evidence.
Decision effect	Absent implementation supports intent only.

APS-022 Effectiveness by condition

Safeguard effectiveness shall be concluded only for validated scope, conditions and period.

Effectiveness is contextual.

Dimension	Requirement
Application	Define trigger, capacity, timing, state and failure modes.
Prohibited interpretation	Generalising one successful test to all conditions.
Minimum evidence	Representative test and operating evidence.
Decision effect	Outside tested conditions remains unproven.

APS-023 Safeguard dependency

Safeguards shall be assessed for power, sensing, control, identity, operator and supplier dependencies.

A safeguard can fail through the same cause it manages.

Dimension	Requirement
Application	Model dependencies and common mode.
Prohibited interpretation	Assuming safeguards are independent.
Minimum evidence	Architecture, configuration and failure tests.
Decision effect	Shared dependency may invalidate barrier claim.

APS-024 Defence diversity

Multiple safeguards strengthen assurance only when mechanisms and dependencies are meaningfully diverse.

Layer count is not defence depth.

Dimension	Requirement
Application	Define diversity and correlated bypass.

Dimension	Requirement
Prohibited interpretation	Counting duplicate policies as independent layers.
Minimum evidence	Design and scenario evidence.
Decision effect	No independence claim without evidence.

APS-025 Residual-path disclosure

Material paths remaining after safeguards shall be disclosed.

Remediation existence does not prove closure.

Dimension	Requirement
Application	Reanalyse alternate and residual paths.
Prohibited interpretation	Reporting only planned control.
Minimum evidence	Post-control analysis and retest.
Decision effect	Residual exposure informs conditions and follow-up.

Part VIII - Evidence and confidence principles

APS-026 Evidence before conclusion

Every material conclusion shall be supported by evidence appropriate to claim type.

Evidence is the basis, not an appendix.

Dimension	Requirement
Application	Match design, implementation and effectiveness claims to corresponding evidence.
Prohibited interpretation	Policy alone for operating effectiveness.
Minimum evidence	Assertion-specific evidence bundle.
Decision effect	Unsupported claim remains UNKNOWN or intent.

APS-027 Evidence provenance

Evidence shall retain source, collection, time, scope, integrity and transformation.

Without provenance evidence cannot be reviewed.

Dimension	Requirement
Application	Use stable IDs and custody records.
Prohibited interpretation	Detached screenshots without context.
Minimum evidence	Evidence manifest and provenance chain.
Decision effect	Material provenance gap limits use.

APS-028 Evidence sufficiency

Sufficiency depends on relevance, reliability, currency, coverage, corroboration and representativeness.

Quantity alone does not create assurance.

Dimension	Requirement
Application	Review each assertion and common-source dependence.
Prohibited interpretation	Averaging evidence grades.
Minimum evidence	Sufficiency rationale.
Decision effect	Insufficient evidence caps the claim.

APS-029 Evidence currency

Evidence remains valid only while representative of the assessed state.

Time and change can invalidate evidence.

Dimension	Requirement
Application	Define expiry and change triggers.

Dimension	Requirement
Prohibited interpretation	Universal fixed validity without change review.
Minimum evidence	Timestamp, configuration and change records.
Decision effect	Stale evidence triggers limitation or reassessment.

APS-030 Assertion-specific confidence

Confidence shall attach to a precise assertion, not a whole report.

Claims in one document may have unequal support.

Dimension	Requirement
Application	State rationale and unresolved conditions per assertion.
Prohibited interpretation	One overall confidence label hiding weak claims.
Minimum evidence	Assertion register and review.
Decision effect	Confidence shapes wording, never replaces evidence.

APS-031 Conflict retention

Conflicting evidence shall remain visible until resolved or bounded.

Deleting conflict creates false certainty.

Dimension	Requirement
Application	Record each source and resolution rationale.
Prohibited interpretation	Selecting favourable evidence silently.
Minimum evidence	Conflict and supersession record.
Decision effect	Unresolved conflict limits assurance.

Part IX - UNKNOWN and limitation principles

APS-032 UNKNOWN preservation

UNKNOWN shall remain a first-class state.

Transparency is safer than fabricated completeness.

Dimension	Requirement
Application	Record what is unknown, materiality, owner and resolution path.
Prohibited interpretation	Encoding UNKNOWN as zero, pass, fail or N/A.
Minimum evidence	Unknown register.
Decision effect	Critical UNKNOWN may cap or block assurance.

APS-033 No assumed favourable state

Absence of adverse evidence shall not prove effectiveness or resilience.

Not finding is not proving.

Dimension	Requirement
Application	Distinguish not assessed, not tested, inconclusive and N/A.
Prohibited interpretation	Assumed pass where evidence is unavailable.
Minimum evidence	Coverage and state records.
Decision effect	No positive claim without affirmative support.

APS-034 Limitation transparency

Material exclusions, sampling, stale evidence and access constraints shall be prominent.

Decision makers need to know what is not covered.

Dimension	Requirement
Application	State limitations beside conclusions and in exports.
Prohibited interpretation	Burying critical limits in appendices.
Minimum evidence	Limitation register.
Decision effect	Limitations shape wording and validity.

APS-035 Not-applicable discipline

N/A requires approved rationale that no relevant subject or outcome exists.

N/A must not remove hard or untested requirements.

Dimension	Requirement
Application	Name criterion, rationale and approver.
Prohibited interpretation	Using N/A because evidence is unavailable.

Dimension	Requirement
Minimum evidence	Applicability decision.
Decision effect	Improper N/A becomes UNKNOWN or gap.

Part X - Critical-gate principles

APS-036 Critical-gate protection

Critical prerequisites shall be evaluated separately from scores and maturity.

Some conditions are indispensable.

Dimension	Requirement
Application	Define gate, failure/unknown criteria, evidence floor and consequence.
Prohibited interpretation	Averaging a failed gate into a good score.
Minimum evidence	Gate record and evidence.
Decision effect	Failed or critical-UNKNOWN gate limits or blocks assurance.

APS-037 Non-compensation

Strong performance in one area shall not compensate for a failed prerequisite or catastrophic scenario.

Substitution hides fragility.

Dimension	Requirement
Application	Use non-compensating logic for safety, continuity, containment, recovery and evidence.
Prohibited interpretation	Points trading across unrelated domains.
Minimum evidence	Gate and outcome analysis.
Decision effect	Decision follows the most restrictive applicable condition.

APS-038 Gate proportionality

Critical gates shall be capability-specific, not a universal static list.

Different profiles have different indispensable conditions.

Dimension	Requirement
Application	Derive gates from outcomes, dependencies and scenarios.
Prohibited interpretation	Declaring every control a gate.
Minimum evidence	Capability profile and rationale.
Decision effect	Improper gate design invalidates score/certification use.

APS-039 Gate closure

A critical gate closes only with corrective evidence and required independent review.

Planned action is not closure.

Dimension	Requirement
Application	Retest condition and residual paths.
Prohibited interpretation	Closure by management promise.

Dimension	Requirement
Minimum evidence	Remediation and retest evidence.
Decision effect	Open gate remains visible in claims.

Part XI - Cross-domain and graph-native principles

APS-040 System-of-systems assurance

AIDCF shall reason across all material domains as one capability-supporting system.

Local conformance does not prove end-to-end outcome.

Dimension	Requirement
Application	Use integrated dependency and scenario views.
Prohibited interpretation	End-to-end decision from siloed checklists only.
Minimum evidence	Cross-domain graph and scenarios.
Decision effect	Missing interaction limits system assurance.

APS-041 Graph-native traceability

Material entities, relations, assertions, paths and decisions should be versioned graph objects.

Graph form makes lineage and propagation explicit.

Dimension	Requirement
Application	Use canonical IDs, evidence and snapshots.
Prohibited interpretation	Treating graph connection as causality or authority.
Minimum evidence	Conformant exchange records.
Decision effect	Graph supports reasoning but proves nothing alone.

APS-042 Federated assurance

AIDCF and AI Trust Graph conclusions shall remain native while contributing to an integrated decision package.

Integration must not collapse semantics.

Dimension	Requirement
Application	Use approved bridge assertions and native snapshots.
Prohibited interpretation	Merging distinct findings into one unsupported verdict.
Minimum evidence	Artifact 005 package.
Decision effect	Integrated decision states separate limitations.

APS-043 Human approval of inference

Automation may propose content; material facts and decisions require accountable review.

Model confidence cannot confer authority.

Dimension	Requirement
Application	Store candidate state, source and reviewer outcome.
Prohibited interpretation	Automatic conversion of inference to fact.

Dimension	Requirement
Minimum evidence	Inference and review log.
Decision effect	Unreviewed inference cannot support final assurance.

Part XII - Decision, scoring, maturity and reporting principles

APS-044 Bounded decision

Every decision shall state subject, question, scope, criteria, evidence, states, period, limitations and owner.

Bounds prevent misleading transfer.

Dimension	Requirement
Application	Use canonical decision record.
Prohibited interpretation	Unqualified statements such as fully assured.
Minimum evidence	Complete decision package.
Decision effect	Missing bounds make decision provisional or invalid.

APS-045 Scoring restraint

Scores may aid prioritisation but shall not create facts, hide uncertainty or override gates.

Numbers can overstate support.

Dimension	Requirement
Application	Publish formula, inputs, coverage and sensitivity.
Prohibited interpretation	Opaque overall score.
Minimum evidence	Transparent scoring and gate record.
Decision effect	Evidence and gates govern wording.

APS-046 Maturity separation

Maturity shall remain distinct from effectiveness, evidence, risk and assurance.

Mature process can still fail a specific outcome.

Dimension	Requirement
Application	Use cumulative criteria and separate gates.
Prohibited interpretation	Inferring resilience from maturity.
Minimum evidence	Maturity evidence and rule-based decision.
Decision effect	Maturity contextualises, not overrides.

APS-047 Reporting integrity

Reports shall give findings, limits, UNKNOWN, conflict, coverage and gate status appropriate prominence.

Selective presentation harms decisions.

Dimension	Requirement
Application	Provide executive and technical traceability.

Dimension	Requirement
Prohibited interpretation	Green dashboards hiding critical limits.
Minimum evidence	Release and traceability review.
Decision effect	Misleading presentation is nonconformance.

APS-048 No inherited assurance

External certification or vendor validation may be evidence but shall not automatically transfer.

Scope, conditions and time differ.

Dimension	Requirement
Application	Assess relevance, currency and coverage.
Prohibited interpretation	Automatic pass by certification inheritance.
Minimum evidence	Third-party report and applicability analysis.
Decision effect	Unsupported inheritance remains a limitation.

Part XIII - Independence, competence and ethics

APS-049 Impartiality

Conclusions shall be protected from commercial, managerial, political and vendor pressure.

Credibility depends on objective interpretation.

Dimension	Requirement
Application	Declare conflicts and separate evaluation, review and decision where required.
Prohibited interpretation	Outcome-contingent conclusion.
Minimum evidence	Conflict register and independent review.
Decision effect	Unmanageable conflict blocks decision.

APS-050 Competence

Assessment shall use collectively competent engineering, AI infrastructure, operations, security and assurance professionals.

No single discipline covers AIDCF.

Dimension	Requirement
Application	Define competence by scope and use specialists.
Prohibited interpretation	General audit competence for licensed engineering conclusions.
Minimum evidence	Competence matrix and assignments.
Decision effect	Gap limits scope or requires specialist review.

APS-051 Professional scepticism

Assessors shall test whether evidence supports claims and consider alternatives.

Assurance is challenge, not confirmation.

Dimension	Requirement
Application	Seek corroboration, exceptions and failed cases.
Prohibited interpretation	Assuming title equals evidence.
Minimum evidence	Review notes and samples.
Decision effect	Unchallenged material claim cannot support high assurance.

APS-052 Confidentiality and minimisation

Collection and reporting shall be authorised, necessary, proportionate and protected.

Assurance does not justify overcollection.

Dimension	Requirement
Application	Apply classification, access, retention and lawful handling.

Dimension	Requirement
Prohibited interpretation	Publishing live sensitive paths.
Minimum evidence	Authorisation and evidence inventory.
Decision effect	Improper material is escalated and may be excluded.

APS-053 Accountability

Owners retain operational responsibility; assessors retain method and conclusion responsibility.

Assurance informs but does not transfer accountability.

Dimension	Requirement
Application	Name owner, assessor, reviewer and decision owner.
Prohibited interpretation	Using assessor opinion to avoid owner decision.
Minimum evidence	Signed role and decision record.
Decision effect	Unclear authority prevents final disposition.

APS-054 Appeal and challenge

Material findings and decisions shall permit controlled challenge.

Fair challenge improves quality.

Dimension	Requirement
Application	Record grounds, evidence, reviewer independence and outcome.
Prohibited interpretation	Informal deletion of disputed findings.
Minimum evidence	Appeal record.
Decision effect	Decision status remains explicit until resolved.

Part XIV - Lifecycle and continuing assurance

APS-055 Temporal validity

Assurance remains valid only for declared snapshot, period and conditions.

Systems, evidence and dependencies change.

Dimension	Requirement
Application	State effective and expiry conditions; preserve history.
Prohibited interpretation	Perpetual validity claim.
Minimum evidence	Snapshot and validity metadata.
Decision effect	Expired conclusion cannot be current.

APS-056 Material-change reassessment

Changes affecting outcomes, dependencies, capacity, state, safeguards or evidence shall trigger impact review.

Prior decisions cannot survive unexamined change.

Dimension	Requirement
Application	Define triggers and affected assertions.
Prohibited interpretation	Updating report date without reassessment.
Minimum evidence	Change and impact record.
Decision effect	Affected conclusions become limited, suspended or superseded.

APS-057 Incident learning

Incidents, tests and near misses shall feed back into scenarios, controls and methods.

Observed outcomes test assumptions.

Dimension	Requirement
Application	Compare event to model and update records.
Prohibited interpretation	Closing incident without methodology feedback.
Minimum evidence	Incident and change record.
Decision effect	Recurring weakness may trigger broader review.

APS-058 Continuing-assurance restraint

Continuous assurance may be claimed only for monitored scope, signals, freshness and decision logic actually implemented.

Telemetry does not create universal assurance.

Dimension	Requirement
Application	State monitored objects, blind spots, cadence and human review.

Dimension	Requirement
Prohibited interpretation	Calling periodic collection continuous assurance.
Minimum evidence	Monitoring design and operation.
Decision effect	Unmonitored conditions remain point-in-time or UNKNOWN.

Part XV - Claim language and assurance decision model

8. Decision statuses

Status	Meaning
NOT ASSESSED	No relevant activity; no conclusion.
UNKNOWN	Evidence absent, insufficient or conflicting.
INCONCLUSIVE	Activity occurred but cannot support determination.
ADVERSE	Evidence demonstrates material unmet condition.
LIMITED	Some claims supported, with material limitations.
CONDITIONAL	Conclusion depends on named continuing conditions.
REASONABLY ASSURED	Sufficient appropriate evidence supports bounded claim and gates.
SUSPENDED	Material change, gate failure or evidence expiry prevents reliance.
WITHDRAWN	Decision authority removed the conclusion.
SUPERSEDED	Newer decision replaces prior current state.

9. Assurance decision record

Field	Required content
decisionId	Stable, versioned identifier.
decisionQuestion	Exact decision supported.
subject/outcomes	Capability Instance and Service Outcomes.
scope/boundary	Inclusions, exclusions, states and period.
criteria/methods	Artifacts, profiles, tests and samples.
evidence/coverage	Manifest, sufficiency and coverage.
unknowns/conflicts	Material unresolved items.
criticalGates	Applicable gate statuses.
findings/residualExposure	Material results and remaining paths.
status/limitations	Controlled status, conditions and restrictions.
validity/triggers	Expiry and reassessment.
roles	Assessor, reviewer and decision owner.
snapshot	Machine-readable reference where applicable.

10. Claim language

Avoid	Use instead
Fully compliant	Assessed against named criteria within stated scope; compliance not determined unless separately authorised.
Zero risk	No material adverse condition identified within tested scope; residual and UNKNOWN remain.

Avoid	Use instead
Fail-proof	Validated for specified scenario and conditions.
Certified by AIDCF	Assessed against named AIDCF artifact/profile, unless an approved scheme applies.
Continuously assured	Monitored for named signals, scope, cadence and review process.
All dependencies identified	No additional material dependency identified within documented discovery coverage.
Control effective	Evidence supports effectiveness for tested scope, period and conditions.
Redundant	Alternative exists; independence and common-mode status stated separately.

11. Assurance ceilings

Constraint	Ceiling
No Capability Instance/Outcome	No capability assurance.
No current boundary/architecture	Descriptive or provisional only.
E0-E1 for material claim	UNKNOWN/candidate only.
Document-only evidence	Design intent at most.
No representative operating/test evidence	No operating-effectiveness claim.
Critical UNKNOWN	Limited, conditional or blocked.
Failed critical gate	Adverse, limited or blocked.
Unassessed degraded/recovery state	No resilience claim for state.
Material change/stale evidence	Suspend or reassess.

Part XVI - Conformance and enforcement

12. Conformance levels

Level	Name	Expectation
P0	Aware	Uses vocabulary without formal claim.
P1	Declared	Identifies applicable principles and deviations.
P2	Assessment conformant	Applies evidence, UNKNOWN, gate, independence and decision rules.
P3	Artifact conformant	Maps downstream requirements to APS principles.
P4	Tool conformant	Preserves semantics, validation, audit and human approval.
P5	Scheme conformant	Demonstrates impartiality, review, surveillance and controlled claims.

13. Nonconformance

Class	Examples	Response
Critical	Fabricated evidence, hidden gate failure, unsafe test, suppressed conflict, misleading certificate.	Stop claim, preserve evidence, escalate and independently review.
Major	Unsupported effectiveness, UNKNOWN converted to pass, boundary omission, unmanaged conflict.	Withdraw/limit and remediate before release.
Minor	Non-material metadata or traceability defect.	Correct under quality review.
Observation	Improvement opportunity without breach.	Track improvement.

14. Exception rule

No exception may waive a constitutional invariant. An alternative may vary implementation only where it demonstrably preserves principle intent and does not weaken safety, evidence, transparency, independence or decision integrity.

Exception field	Requirement
principle/rule	Exact item affected.
reason	Why standard approach is impracticable.
alternative	How intent remains satisfied.
impact	Evidence and decision effect.
owner/approver	Authorised and independent roles.
duration	Time-bounded validity.
monitoring	Conditions and reassessment.
disclosure	Report and machine-readable placement.
closure	Evidence of end state.

Part XVII - Machine-readable constitutional profile

15. Entities and relationships

Entity	Purpose
AssurancePrinciple	Canonical constitutional rule.
PrincipleObligation	Required application.
Prohibition	Disallowed practice.
AssuranceClaim	Precise proposition.
CriticalGate	Non-compensating prerequisite.
EvidenceRequirement	Minimum support.
UnknownRecord	Unresolved information and effect.
Limitation	Restriction on interpretation.
AssuranceDecision	Bounded authorised conclusion.
ExceptionRecord	Approved equivalent treatment.

Entity	Purpose
Nonconformance	Principle breach.
ReassessmentTrigger	Condition requiring review.

Relationship	Meaning
GOVERNS	Principle constrains artifact, claim or process.
REQUIRES	Principle mandates obligation.
PROHIBITS	Principle disallows practice.
SUPPORTED_BY	Claim/decision linked to evidence.
LIMITED_BY	Claim constrained by limitation/UNKNOWN.
BLOCKED_BY	Gate prevents conclusion.
REVIEWED_BY	Accountable review.
SUPERSEDES	New version replaces without erasing history.
TRIGGERS_REASSESSMENT_OF	Change affects prior decision.
CONFORMS_TO	Artifact/tool declares tested conformance.

16. Publication package

File	Purpose
aidcf-aps-v1.0.owl	Formal constitutional ontology.
aidcf-aps-v1.0.ttl	RDF/Turtle serialisation.
aidcf-aps-context-v1.0.jsonld	JSON-LD context.
aidcf-aps-schema-v1.0.json	Decision, gate and exception schemas.
aidcf-aps-shapes-v1.0.ttl	Validation shapes.
principles-v1.0.csv	Principle registry.
critical-gates-v1.0.csv	Gate catalogue.
claim-language-v1.0.csv	Controlled wording.
conformance-tests/	Positive/negative fixtures.
CHANGELOG.md / MIGRATION.md	Release history and transition guidance.

17. Validation rules

Rule	Validation statement
APS-VAL-001	Every downstream requirement maps to at least one principle.
APS-VAL-002	Every final decision identifies capability and outcomes.
APS-VAL-003	Every decision has scope, state, period, evidence and limitations.
APS-VAL-004	Every material assertion has evidence or explicit UNKNOWN.
APS-VAL-005	Every applicable gate has status and evidence.

Rule	Validation statement
APS-VAL-006	No score or maturity overrides a gate.
APS-VAL-007	No UNKNOWN becomes favourable numeric/boolean state.
APS-VAL-008	Every effectiveness claim identifies tested conditions.
APS-VAL-009	Every material change links affected decisions.
APS-VAL-010	Every AI-generated item remains candidate until reviewed.
APS-VAL-011	Every exception is time-bound and preserves intent.
APS-VAL-012	Every nonconformance has disposition.
APS-VAL-013	Every integrated package preserves native decisions.
APS-VAL-014	Every published claim uses controlled status.
APS-VAL-015	Every superseded decision remains traceable.

Part XVIII - Worked applications and anti-patterns

18. AI training capability

Step	Application
Capability	Define training capacity, completion, checkpoint and recovery outcomes.
Boundary	Declare halls, clusters, storage, fabric, operators and providers.
Dependencies	Trace power, liquid cooling, management, identity, scheduler, storage, spares and OEM.
Failure	Analyse cooling, power, metadata, congestion and common-mode firmware scenarios.
Evidence	Separate design, as-built, commissioned, operating and tested evidence.
Gates	Thermal, power, checkpoint, recovery and observability gates.
Decision	Bounded readiness with UNKNOWN, residual paths and triggers.

19. Colocation capability

Issue	Treatment
Provider certification	Evidence only; no automatic inheritance.
Shared responsibility	Owner/operator/custodian by function and state.
Excluded provider system	Visible external dependency with confidence limit.
Boundary	Physical, network, identity, operational and evidence crossings.
Testing constraint	Authorised safe validation and disclosed limitation.
Critical UNKNOWN	Evidence-inaccessible shared dependency may cap assurance.

20. Continuous monitoring

Question	Required answer
What is monitored?	Named entities, relations, safeguards and metrics.
What is not?	Blind spots and manual dependencies.
How current?	Collection/review cadence and latency.
What changes a decision?	Threshold, gate and human-review logic.
Coverage?	Assets, states, sites, periods and scenarios.
Point-in-time elements?	Tests, inspections, studies and interviews.
Permitted claim	Only monitored evidenced scope.

21. Anti-patterns

ID	Anti-pattern
APAT-01	Checklist completion presented as capability assurance.
APAT-02	Facility rating inherited by every AI workload.

ID	Anti-pattern
APAT-03	Average score masks critical power, cooling, recovery or evidence gate.
APAT-04	Redundancy claimed from asset count.
APAT-05	Policy used to claim technical effectiveness.
APAT-06	UNKNOWN removed to complete dashboard.
APAT-07	Normal-state test generalised to recovery.
APAT-08	External dependency omitted due to contract.
APAT-09	Graph connection treated as authority or causality.
APAT-10	Tool finding released without human approval.
APAT-11	Maturity used as proxy for resilience.
APAT-12	Planned remediation treated as gate closure.
APAT-13	Report treated as current after material change.
APAT-14	Sensitive path published without protection.
APAT-15	Certification claim without approved scheme.

Appendix A - Principle registry

ID	Principle
APS-001	Capability-first assurance
APS-002	Outcome alignment
APS-003	Function before product
APS-004	Outcome-centric materiality
APS-005	Explicit boundary
APS-006	Operating-envelope discipline
APS-007	State-specific assurance
APS-008	Reality over intent
APS-009	Proportionate depth
APS-010	Dependency visibility
APS-011	External dependency continuity
APS-012	Shared-dependency scrutiny
APS-013	Concentration awareness
APS-014	Dependency ownership
APS-015	Failure-chain analysis
APS-016	Cross-domain propagation
APS-017	Compound conditions
APS-018	Resilience lifecycle
APS-019	Recovery evidence
APS-020	Safe validation
APS-021	Safeguard reality
APS-022	Effectiveness by condition
APS-023	Safeguard dependency
APS-024	Defence diversity
APS-025	Residual-path disclosure
APS-026	Evidence before conclusion
APS-027	Evidence provenance
APS-028	Evidence sufficiency
APS-029	Evidence currency
APS-030	Assertion-specific confidence
APS-031	Conflict retention
APS-032	UNKNOWN preservation
APS-033	No assumed favourable state
APS-034	Limitation transparency

ID	Principle
APS-035	Not-applicable discipline
APS-036	Critical-gate protection
APS-037	Non-compensation
APS-038	Gate proportionality
APS-039	Gate closure
APS-040	System-of-systems assurance
APS-041	Graph-native traceability
APS-042	Federated assurance
APS-043	Human approval of inference
APS-044	Bounded decision
APS-045	Scoring restraint
APS-046	Maturity separation
APS-047	Reporting integrity
APS-048	No inherited assurance
APS-049	Impartiality
APS-050	Competence
APS-051	Professional scepticism
APS-052	Confidentiality and minimisation
APS-053	Accountability
APS-054	Appeal and challenge
APS-055	Temporal validity
APS-056	Material-change reassessment
APS-057	Incident learning
APS-058	Continuing-assurance restraint

Appendix B - Baseline critical-gate catalogue

This is a candidate baseline. Profiles determine applicability, detailed criterion and evidence floor. It is not a universal pass/fail list.

ID	Gate
CG-01	Capability and outcome definition
CG-02	Boundary and architecture integrity
CG-03	Critical dependency visibility
CG-04	Life safety and hazardous energy
CG-05	Resilient electrical supply
CG-06	Thermal continuity

ID	Gate
CG-07	Critical control-system integrity
CG-08	Compute, storage and network functional integrity
CG-09	Failure containment
CG-10	Recovery readiness
CG-11	Evidence sufficiency
CG-12	Critical UNKNOWN
CG-13	Responsibility and authority
CG-14	Change control
CG-15	Integrated digital authority over infrastructure

Gate record schema

Field	Requirement
gateId	Stable ID and profile/version.
subject/outcome	Capability and Outcome protected.
criterion	Testable prerequisite.
applicability	Why it applies.
failure/unknown condition	When gate fails or remains UNKNOWN.
evidence floor	Minimum support.
methods	Analysis, inspection, test and review.
status	Pass, Fail, UNKNOWN or approved N/A.
decision effect	Block, cap, condition or escalate.
owner/reviewer	Accountable roles.
closure	Remediation and retest.

Appendix C - UNKNOWN and limitation schemas

UNKNOWN field	Content
unknownId	Stable ID.
subject/attribute	What is unresolved.
reason	Absent, inaccessible, stale, conflicting or not tested.
potential consequence	Why it matters.
criticality	Criticality and rationale.
evidence needed	Resolution requirement.
owner	Owner if known.
decision effect	Limit, condition, block or suspend.

UNKNOWN field	Content
status/history	Open, resolved, accepted or superseded.

Limitation field	Content
limitationId	Stable ID.
affected claim	Claim/decision affected.
type	Scope, evidence, state, method, access, time, competence or legal.
description	Precise restriction.
materiality	Decision effect.
owner/treatment	Resolution owner and treatment.
visibility	Where disclosed.
trigger	Review/expiry condition.

Appendix D - Conformance checklist

Check	Requirement
D-01	Capability Instance and Outcomes defined.
D-02	Boundary, envelope, states and period declared.
D-03	Dependencies and providers visible.
D-04	Failure and recovery paths analysed cross-domain.
D-05	Safeguards validated by condition.
D-06	Evidence linked to assertions with provenance.
D-07	UNKNOWN, conflicts and limitations preserved.
D-08	Critical gates defined and non-compensating.
D-09	Scores/maturity do not override gates.
D-10	Decision bounded and controlled.
D-11	Competence, impartiality and review demonstrated.
D-12	Change and reassessment triggers stated.
D-13	Tool output retains human approval.
D-14	Sensitive evidence protected.
D-15	Exceptions and nonconformance controlled.

Appendix E - Review and acceptance record

Review role	Status	Focus
Methodology author	Draft completed	Constitutional coherence.

Review role	Status	Focus
AIDCF ontology reviewer	Pending	Artifact 003 consistency.
Failure/resilience reviewer	Pending	Artifact 004 consistency.
AI Trust Graph integration reviewer	Pending	Artifact 005 consistency.
Electrical engineer	Pending	Power gate validity.
Mechanical/cooling engineer	Pending	Thermal gate validity.
Compute/storage/network reviewer	Pending	AI infrastructure outcomes.
Operations/OT reviewer	Pending	State, automation and authority.
Assurance reviewer	Pending	Evidence, independence and claims.
Certification/accreditation reviewer	Pending	Scheme implications.
Security/privacy reviewer	Pending	Evidence/path protection.
Legal/IP/publication reviewer	Pending	Rights and claims.
Independent practitioner	Pending	Reproducibility.
Public release	Not approved	All gates closed.

Acceptance criteria

- Downstream artifacts can map to this standard without contradiction.
- Evidence, UNKNOWN, gates, independence and bounded decisions are enforceable.
- No principle implies legal compliance or engineering approval by itself.
- Machine-readable implementation does not change meaning.
- Independent multidisciplinary and publication review is completed.

Change record

Version	Status	Summary
0.1	Architecture draft	Constitutional role and principle families.
0.5	Integrated draft	Decision, gate, ethics and conformance models.
1.0	Controlled-review draft	Comprehensive baseline for downstream artifacts.

Final constitutional doctrine

AIDCF may express confidence only where the decision question is clear, the capability and outcomes are defined, dependencies and failure paths are sufficiently understood, safeguards are evidenced within relevant conditions, uncertainty remains visible, critical gates are satisfied, competent and impartial review has occurred, and the conclusion is bounded by scope, state, time and evidence. Where these conditions are not met, AIDCF shall reduce, condition, suspend or withhold assurance rather than manufacture certainty.